

LAB INSTRUCTION GUIDE

Static VXLAN Configuration

on Cisco Nexus 9000v

Platform: **Cisco Nexus 9000v (NX-OS 9.x)** Environment: **EVE-NG**

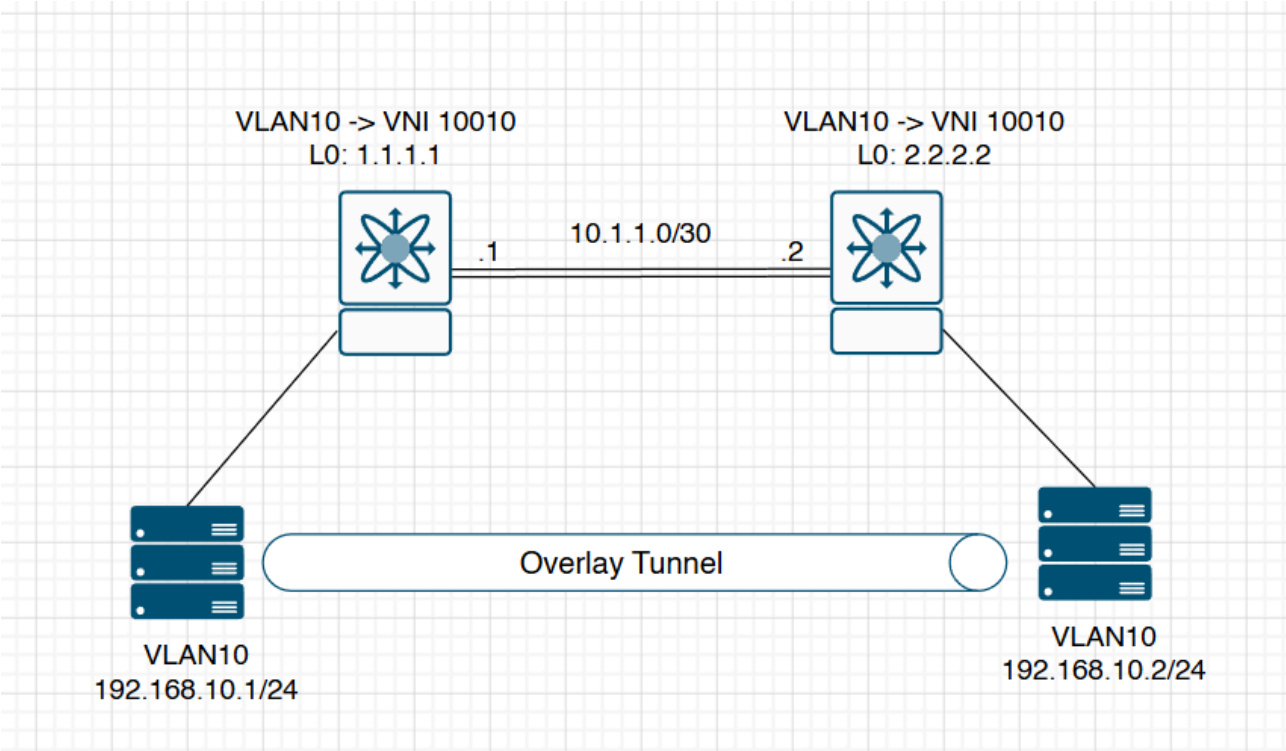
Overview

This guide provides a complete, battle-tested procedure for deploying a Static VXLAN overlay on Cisco Nexus 9000v switches running NX-OS 9.x within an EVE-NG virtual lab environment. Each phase has been validated to account for the undocumented TCAM constraints and architectural requirements specific to this platform.

Lab Objectives	Configure a functional Layer 2 VXLAN tunnel (ingress replication, static peer) between two Nexus 9K leaf switches, verified by live traffic.
Estimated Time	< 15 minutes (5 minutes for experienced engineers)
Prerequisites	EVE-NG instance with Nexus 9000v images loaded; two switches provisioned with console access; VPCS or Linux end-hosts configured.

Lab Topology

The following diagram shows the full topology. Two Nexus 9K leaf switches act as VTEPs (VXLAN Tunnel Endpoints) connected back-to-back on Ethernet1/1 (underlay). End-host VPCs attach on Ethernet1/2 and reside in VLAN 10 on both sides.



Address Plan

Device	Interface	IP Address	Purpose
Nexus-9K-01	Loopback0	1.1.1.1/32	VTEP Source IP
Nexus-9K-01	Ethernet1/1	10.1.1.1/30	Underlay P2P Link
Nexus-9K-01	VLAN 10 / Eth1/2	—	Access to VPC1
Nexus-9K-02	Loopback0	2.2.2.2/32	VTEP Source IP
Nexus-9K-02	Ethernet1/1	10.1.1.2/30	Underlay P2P Link
Nexus-9K-02	VLAN 10 / Eth1/2	—	Access to VPC2
VPC1	eth0	192.168.10.1/24	End-host
VPC2	eth0	192.168.10.2/24	End-host

Phase 0 Virtual Hardware Preparation — TCAM Carving

Before enabling any NX-OS features, the virtual TCAM regions must be carved to their correct sizes. Skipping this step causes ACLQOS_FAILED errors at boot and unpredictable feature failures.

IMPORTANT: This step is required on BOTH switches. The switches must be reloaded after applying the TCAM configuration for changes to take effect.

Apply the following on Nexus-9K-01 and Nexus-9K-02:

```
configure terminal
hardware access-list tcam region ing-copp 256
hardware access-list tcam region qos-ingress 256
copy run start
reload
```

Wait for both switches to complete the reboot cycle and return to the NX-OS prompt before proceeding to Phase 1.

Phase 1 Underlay Configuration — OSPF Routing

The underlay provides IP reachability between the two VTEP Loopback addresses. OSPF is used as the routing protocol. Both the point-to-point link and the Loopback interface must be placed into OSPF Area 0.

IMPORTANT: The OSPF process must be created globally before it can be applied to interfaces. The Loopback address is the VTEP source IP — it must be reachable from the remote switch.

Nexus-9K-01

```
configure terminal
feature ospf
router ospf 1

interface Ethernet1/1
no switchport
ip address 10.1.1.1/30
ip router ospf 1 area 0.0.0.0
no shutdown

interface Loopback0
ip address 1.1.1.1/32
ip router ospf 1 area 0.0.0.0
no shutdown
```

Nexus-9K-02

```
configure terminal
feature ospf
router ospf 1

interface Ethernet1/1
 no switchport
 ip address 10.1.1.2/30
 ip router ospf 1 area 0.0.0.0
 no shutdown

interface Loopback0
 ip address 2.2.2.2/32
 ip router ospf 1 area 0.0.0.0
 no shutdown
```

Verification

Run the following command on each switch and confirm the remote Loopback address appears in the OSPF routing table:

```
show ip route ospf
```

Expected output: Nexus-9K-01 should see 2.2.2.2/32 via OSPF. Nexus-9K-02 should see 1.1.1.1/32 via OSPF. Do not advance to Phase 2 until both routes are present.

Phase 2 Overlay Configuration — VNI & Ingress Replication

This phase builds the VXLAN overlay. VLAN 10 is mapped to VNI 10010 on both switches. Ingress replication is used instead of multicast, so each switch must be explicitly configured with the remote VTEP's Loopback IP as a static peer.

NOTE: In NX-OS 9.x, the ingress-replication configuration must be nested under the member vni hierarchy inside the NVE interface. The command syntax differs from older NX-OS releases.

Nexus-9K-01

```
configure terminal
feature nv overlay
feature vn-segment-vlan-based

vlan 10
 vn-segment 10010

interface nve1
 no shutdown
```

```
source-interface Loopback0
member vni 10010
  ingress-replication protocol static
  peer-ip 2.2.2.2
```

Nexus-9K-02

```
configure terminal
feature nv overlay
feature vn-segment-vlan-based

vlan 10
  vn-segment 10010

interface nve1
  no shutdown
  source-interface Loopback0
  member vni 10010
    ingress-replication protocol static
    peer-ip 1.1.1.1
```

Phase 3 Access Layer — Active Attachment Circuit

NX-OS hardware optimization holds the NVE interface and VNI in a Down state until at least one physical access port assigned to the mapped VLAN comes up. This is known as the Active Attachment Circuit requirement.

IMPORTANT: The NVE interface will NOT activate until an access port in VLAN 10 reaches an Up/Up state. VPC1 and VPC2 must be powered on in the EVE-NG topology.

Both Switches — Access Port Configuration

```
configure terminal
interface Ethernet1/2
  switchport
  switchport access vlan 10
  no shutdown
```

After applying the configuration, power on VPC1 and VPC2 in the EVE-NG topology and verify the switchport line protocol reaches Up/Up before continuing.

Phase 4 Data Plane Verification

Because this deployment uses static ingress replication (no BGP EVPN control plane), the switch does not establish peer adjacency until data traffic forces the ARP and MAC learning process. A single ping from VPC1 to VPC2 triggers this.

Step 1 — Trigger the Tunnel

From VPC1, issue a ping to VPC2:

```
ping 192.168.10.2
```

EXPECTED BEHAVIOUR: The first packet will fail (host unreachable or timeout). The switch must encapsulate the ARP broadcast and dynamically learn the remote MAC address. The second and subsequent packets will succeed.

Step 2 — Verification Commands

Run the following commands on either switch to confirm the VXLAN tunnel is fully operational:

Command	Expected Result
show nve vni	State: Up Mode: DP Multicast-group: UnicastStatic
show nve peers	Remote Loopback IP listed with State: Up
show mac address-table vlan 10	Remote MAC address visible with NVE1 as the egress interface

Step 3 — Packet Capture (Optional Confirmation)

For definitive proof of VXLAN encapsulation, run a packet capture on Ethernet1/1 (the underlay interface) and filter for UDP port 4789. The capture should show the original Layer 2 Ethernet frames from the 192.168.10.0/24 subnet encapsulated inside UDP/IP packets between the two Loopback addresses.

Troubleshooting Reference

Symptom	Resolution
ACLQOS_FAILED error at boot	TCAM regions were not carved (Phase 0). Apply TCAM config and reload both switches.
NVE interface stuck in Down state	Ensure Ethernet1/2 is up and assigned to VLAN 10. VPCs must be powered on.
show ip route ospf shows no remote Loopback	Verify Ethernet1/1 is set to no switchport and OSPF is enabled on both interfaces with the correct area.
Ping fails after tunnel appears Up	Trigger the data plane with a first ping (expected to fail). The second ping should succeed as MAC addresses are learned.
show nve peers shows no peers	First ping has not been sent. Send traffic from VPC1 to VPC2 to trigger peer discovery.

Key Architectural Notes

- VXLAN encapsulates Layer 2 frames inside UDP packets (destination port 4789), allowing Layer 2 domains to span Layer 3 boundaries.
- Static ingress replication eliminates the need for multicast or BGP EVPN. Each VTEP unicasts BUM (Broadcast, Unknown Unicast, Multicast) traffic directly to each configured peer.
- The Active Attachment Circuit rule is a hardware optimisation — the NVE interface will not consume TCAM resources until a live access port exists in the mapped VLAN.
- Because there is no control plane, MAC and ARP learning is purely data-plane driven. The first packet in each flow triggers learning.
- VNI 10010 is the VXLAN Network Identifier. All switches participating in the same Layer 2 segment must use the identical VNI value.

Running config

SW1

```
vlan 10
  vn-segment 10010
interface nve1
  no shutdown
  source-interface loopback0
  member vni 10010
    ingress-replication protocol static
    peer-ip 2.2.2.2
interface Ethernet1/1
  no switchport
  ip address 10.1.1.1/30
  ip router ospf 1 area 0.0.0.0
  no shutdown
interface Ethernet1/2
  switchport access vlan 10
interface loopback0
  ip address 1.1.1.1/32
  ip router ospf 1 area 0.0.0.0
router ospf 1
```

SW2

```
vlan 10
  vn-segment 10010
interface nve1
  no shutdown
  source-interface loopback0
  member vni 10010
    ingress-replication protocol static
    peer-ip 1.1.1.1
interface Ethernet1/1
  no switchport
  ip address 10.1.1.2/30
  ip router ospf 1 area 0.0.0.0
  no shutdown
```



```
interface Ethernet1/2
  switchport access vlan 10
interface loopback0
  ip address 2.2.2.2/32
  ip router ospf 1 area 0.0.0.0
router ospf 1
```